

U.G.C. NET Exam COMPUTER (Solved with Explanations)

1. Which of the following is correct for the given poset?
- (a) There exists a greatest element and a least element
 - (b) There exists a greatest element but not a least element
 - (c) There exists a least element but not a greatest element
 - (d) There does not exist a greatest element and a least element

Ans. (d) : A relation R on set A is called partial ordering or partial order if it is reflexive, anti-symmetric and transitive. A set together with a partial ordering R is called a partial ordered set or poset. The poset is denoted as (S, R) .

Relations can be used to order some or all the elements of a set. For instance, the set of natural number is ordered by the relation $\leq$ such that for every ordered pair (x, y) in the relation, natural number x comes before the natural number y unless both are equal. So option fourth is correct.

2. How many ways are there to place 8 indistinguishable balls into four distinguishable bins?
- (a) 70
 - (b) 165
 - (c) 8C_4
 - (d) 8P_4

Ans. (b) : If the question ask about number of ways in which m indistinguishable balls is placed into n distinguishable boxes then it is a question of selection of m boxes out of n boxes.

Two cases may arise :-

- (a) No. of balls $m >$ No. of boxes, n .

Then we have to use selection of boxes where repetition is allowed.

$$\text{No. of ways} = {}^{n+m-1}C_m$$

- (b) No. of balls, $m \leq$ no. of boxes, n

Then we have to use selection of m boxes out of n boxes where no repetition is allowed.

$$\text{No. of ways} = {}^nC_m$$

In this question, $m = 8$, and $n = 4$

Since $m > n$ we have to use selection of n boxes within repetition

$$\begin{aligned}\therefore \text{Required number of ways} &= {}^{n+m-1}C_m \\ &= {}^{4+8-1}C_8 = {}^{11}C_8 = 165\end{aligned}$$

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3. How many bit strings of length ten either start with a 1 bit or end with two bits 00?

- (a) 320
- (b) 480
- (c) 640
- (d) 768

Ans. (c) : The problem can be divided into two parts,

Part 1:- No. of bit strings of length 10 that start with bit 1

So, the strings will be of the form

1 _ _ _ _ _

which means 9 empty places to fill with either 0 or 1, hence 2^9 ways.

Part 2:- No of bit strings of length 10 that end two bit zero.

Now we have already counted the strings the end with two bit zeros in the part 1, but all those strings. Were starting with a bit 1, so we still haven't counted the bit strings that start with a zero and end with two zeros. So such string will be of the form.

0 _ _ _ _ _ 00

which means 7 empty places to fill with either 0 or 1 hence 2^7 ways.

Hence total no of ways = part 1 + part 2

$$\begin{aligned}
 &= 2^9 + 2^7 \\
 &= 2^7 * (2^2 + 1) \\
 &= 128 \times 5 \\
 &= 640
 \end{aligned}$$

4. Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph?

- (a) 6
- (b) 8
- (c) 12
- (d) 20

Ans. (b) : We apply Euler's formula where $r = e - v + 2$ since each vertex has degree 4, the sum of the degree is 24.

By handshaking theorem, $2e = 24$

So $e = 12$

$r = 12 - 6 + 2$

$r = 8$

So we have 8 regions in this graph.

5. For which values of m and n does the complete bipartite graph $K_{m,n}$ have a Hamilton circuit?

- (a) $m \neq n, m, n \geq 2$

- (b) $m \neq n, m, n \geq 3$
- (c) $m = n, m, n \geq 2$
- (d) $m = n, m, n \geq 3$

Ans. (c) : Complete bipartite graph : In this graph vertices are partitioned into two sets such that no vertices are connected in the same set and all vertices of one set are connected to all other vertices of different set. All possible pairs of edges are considered except edges in the same set.

Hamiltonian Circuit $\rightarrow$ In this circuit, every vertex is visited only once. It must start and end at the same vertex.

In complete bipartite graph $K_{m,n}$ when $m = n$, then in that case, it has a Hamiltonian circuit consider $m = n = 3$

So option c. is the correct answer. (option d is also correct but option c is more appropriate since it covers the case $m, n = 2$)

For a bipartite graph $(K_{m,n})$ to be a Hamiltonian graph $m = n$ and $m, n \geq 2$

6. Which of the following is principal conjunctive normal form for $[(p \vee q) \wedge \neg p \rightarrow \neg q]$?

- (a) $p \vee \neg q$
- (b) $p \vee q$
- (c) $\neg p \vee q$
- (d) $\neg p \vee \neg q$

Ans. (a) : Given $(p \vee q) \wedge \neg p \rightarrow \neg q$

The precedence of the used operators are :-

$$\wedge > \vee \rightarrow$$

Therefore, $[(p \vee q) \wedge \neg p \rightarrow \neg q]$

$$\Leftrightarrow [((p \vee q) \wedge \neg p) \rightarrow \neg q]$$

$$\Leftrightarrow [((p \wedge \neg q) \vee (q \wedge \neg p)) \rightarrow \neg q] \text{ -- using distributive law}$$

$$\Leftrightarrow [(p \vee (q \wedge \neg p)) \rightarrow \neg q]$$

$$\Leftrightarrow [(q \wedge \neg p) \rightarrow \neg q]$$

$$\Leftrightarrow [\neg (q \wedge \neg p) \vee \neg q] \text{ -- using demorgan's law}$$

$$\Leftrightarrow [\neg q \vee p \vee \neg q] \Leftrightarrow [p \vee \neg q]$$

7. How many cards must be selected from a standard deck of 52 cards to guarantee that at least three hearts are present among them?

- (a) 9
- (b) 13
- (c) 17
- (d) 42

Ans. (d) : In this the worst case will happen as follows :-

1st 13 * 3 = 39 cards that we pick up turns out to be any one of diamonds or spades or clubs but not hearts.

Now when we pick the next 3 cards they are guaranteed to be hearts ($\because$ we have already picked cards from all other suits)

$\therefore$ We need to pick $13 + 13 + 13 + 3 = 42$ cards in order to guarantee that there are atleast 3 cards of hearts in the selected cards.

8. Match List-I with List-II :

List-I	List-II
(A) $p \rightarrow q$	(i) $\neg(q \rightarrow \neg p)$
(B) $p \vee q$	(ii) $p \wedge \neg q$
(C) $p \wedge q$	(iii) $\neg p \rightarrow q$
(D) $\neg(p \rightarrow q)$	(iv) $\neg p \vee q$

Choose the correct option from those given below :

(a) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 (b) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 (c) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 (d) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Ans. (d) :

List-I	List-I
(A) $p \rightarrow q$	(iv) $\neg p \vee q$
(B) $p \vee q$	(iii) $\neg p \rightarrow q$
(C) $p \wedge q$	(i) $\neg(q \rightarrow \neg p)$
(D) $\neg(p \rightarrow q)$	(ii) $p \wedge \neg q$

(A) $p \rightarrow q = \neg p \vee q$

(B) $p \vee q$

(C) $p \wedge q$

(D) $\neg(p \rightarrow q) = \neg(p \vee q) = p \wedge \neg q$

(i) $\neg(q \rightarrow \neg p) = \neg(\neg q \vee \neg p) = q \wedge p$

(ii) $p \wedge \neg q$

(iii) $\neg p \rightarrow q = p \vee q$

(iv) $\neg p \vee q$

9. Find the zero-one matrix of the transitive closure of the relation given by the matrix A :

$A =$	$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
(a)	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
(b)	$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
(c)	$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
(d)	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Ans. (a) : Let the elements be a, b, c on which the relation is defined.

Then from the matrix we can say that the relation given is

$\{(a, a), (a, c), (b, b), (c, a), (c, b)\}$

Now, what are the minimum number of ordered pairs that we should add to the given relation that it becomes transitive relation ?

Using Warshall's Algorithm we can see that (c, c) and (a, b) are missing.

After adding them to the relation and making the matrix again the matrix will become

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

10. Consider and LPP given as

$$\text{Max } Z = 2x_1 - x_2 + 2x_3$$

Subject to the constraints

$$2x_1 + x_2 \leq 10$$

$$x_1 + 2x_2 - 2x_3 \leq 20$$

$$x_1 + 2x_2 \leq 5$$

$$x_1, x_2, x_3 \geq 0$$

What shall be the solution of the LPP after applying first iteration of the Simplex Method?

(a) $x_1 = \frac{5}{2}, x_2 = 0, x_3 = 0, Z = 5$

(b) $x_1 = 0, x_2 = 0, x_3 = \frac{5}{2}, Z = 5$

(c) $x_1 = 0, x_2 = \frac{5}{2}, x_3 = 0, Z = -\frac{5}{2}$

(d) $x_1 = 0, x_2 = 0, x_3 = 10, Z = 20$

Ans. (b) :

$$\text{Given max } z = 2x_1 - x_2 + 2x_3 + 0S_1 + 0S_2 + 0S_3$$

$$2x_1 + x_2 + s_1 = 10$$

$$x_1 + 2x_2 - 2x_3 + S_2 = 20$$

$$x_1 + 2x_3 + S_3 = 5$$

initial simplex table :

	Cj=	2	-1	2	0	0	0		Theta
CB	Base	x_1	x_2	x_3	S_1	S_2	S_3	b	
0	S_1	2	1	0	1	0	0	10	ND
0	S_2	1	2	-2	0	1	0	20	-10
0	S_3	1	0	2	0	0	1	5	5/2
$Z_j = \sum \{C_B\} x_j$		0	0	0	0	0	0		
$C_j - Z_j$		2	-1	2	0	0	0		

Here in C_j , we have two numbers which are highest (ie. 2 in column 1st and 2 in column 3rd)

So we can choose any column to begin with :

So lets start with column 3rd for convenience.

Hence column 3rd become key column.

Now, we find $\text{Theta} = b/\text{corresponding key column element}$.

Now, choose the lowest positive number including zero in theta and mark if key row. Intersection on key column and key row is key elements key column is X_u. Key row is the minimum value in Theta i.e 1st row. key element is 2($S_1 X_1$) But key element should be 1 so, divide the complete row by 2								
	C_j	2	-1	2	0	0	0	
CB	Base	X_1	X_2	X_3	S_1	S_2	S_3	b
0	S_1	2	1	0	1	0	0	10
0	S_2	0	2	0	0	1	1	15
2	X_3	1/2	0	1	0	0	1/2	5/2
$Z_j = \backslash$ ($\backslash$ sum {C- B}{X- J}\)		0	0	0	0	0	0	5
$C_j=C_j-$ Z_j		2	-1	2	0	0	0	
Hence 1st iteration table is prepared b indicates value of variable which is corresponding in column C_j (only take the give X_i variable). therefore $X_1 = 0$, $X_2 = 0$, $X_3 = 5/2$ b indicated values of $Z_{\max}$ in Row C_j and therefore $Z_{\max} = 5$								

11. Which type of addressing mod, less number of memory references are required?

- (a) Immediate
- (b) Implied
- (c) Register
- (d) Indexed

Ans. (c) : (Dropped Question by NTA) Immediate addressing mode- may or may not need memory reference Implied addressing mode- may or may not need memory reference (Like push, pop or CMA instruction) Register Addressing mode- It does not need memory reference as operands one always in register. Indexed addressing mode- need memory reference.
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12. Suppose that the register A and the register K have the bit configuration. Only the three leftmost bits of A are compared with memory words because K has I's in these positions. Because of its organization, this type of memory is uniquely suited to parallel searches by data association. This type of memory is known as

- (a) RAM
- (b) ROM
- (c) content addressable memory
- (d) secondary memory

Ans. (c) : The time required to find an item stored in memory can be reduced considerably if stored data is identified for access by the content of the data itself rather than by an address. Memory unit associated by content is called an associative memory or content addressable memory (CAM).

CAM is accessed simultaneously and in parallel on the basis of data content rather than specific location. When a word is written in CAM, no address is given. Memory is capable of finding an unused empty slot to store the

Ans. (a) : In general for n variable there are 2^n rows and
Degree of Boolean function = number variables = n
Variable is of Boolean type

Sequence possible of two Boolean = 2^n

Different Boolean function = $\setminus(\{2\}^{\{\{2\}^{\{n\}}\}}\setminus)$

Example :

By taking $n = 2$

A	B	F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	0
1	1	0	1	0	1	0	1	0	1

Different Boolean function = $\setminus(\{2\}^{\{\{2\}^{\{n\}}\}} = \{2$
Therefore 16 Boolean function is possible.

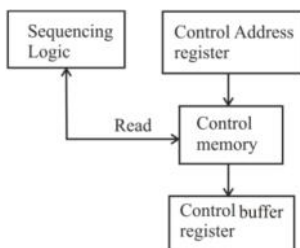
14. The fault can be easily diagnosed in the micro-program control unit using diagnostic tools by maintaining the contents of

- (a) flags and counters
- (b) registers and counters
- (c) flags and registers
- (d) flags, registers and counters

Ans. (d) : Flags, registers and counters-

within the computer, a control unit manages the computer's resources and orchestrates the performance of its functional parts in response to the instructions.

Microprogrammed control unit operates by executing instructions that define the functionality of the control unit. In this, set of micro instructions is stored in the control memory. The control address register contains the address of the next micro instruction to be read. When a micro instruction is read from the control memory, it is transferred to a control buffer register. Next element is sequencing unit that loads the control address register and issues a read command.



word. When a word is to be read from memory, the content of the word is specified, and memory locates all words which matches the specified content and marks them for reading.

13. How many different Boolean functions of degree n are there?

- (a) 2^{2^n}
- (b) $(2^2)^n$
- (c) $2^{2^n} - 1$
- (d) 2^n

2^{2^n} possible functions.

F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}
1	1	1	1	1	1	1	1
0	0	0	0	0	1	1	1
0	0	1	1	1	0	1	1
0	1	0	1	0	1	0	1

$\}^{\{2\}}\}^{\{2\}} = 16\backslash)$

15. What will be the number of states when a MOD-2 counter is followed by a MOD-5 counter?

- (a) 5
- (b) 10
- (c) 15
- (d) 20

Ans. (b) : When mod- x counter is followed by mod-y counter then the resultant counter is always mod-(x * y) counter.

So simply multiply MOD value, answer will be 10.

16. For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to

- (a) non-uniform distribution of requests
- (b) arm starting or stopping inertia
- (c) higher capacity of tracks on the periphery of the platter
- (d) use of unfair arm scheduling policies

Ans. (b) : Whenever head moves from one track to other then its speed and direction changes. Which is nothing but change in motion or the case of inertia.

The positioning time, or random-access time, consists of two parts : the time necessary to move the disk arm to the desired cylinder, called the seek time, and the time necessary for the desired sector to rotate to the disk head, called the rotational latency.

The seek latency is not linearly proportional to the seek distance due to arm starting and stopping inertia. It means seek time is not directly dependent on the seek distance traveled, but also on the accelerational deceleration (on inertia) of actuator arm.

17. The parallel bus arbitration technique uses an external priority encoder and a decoder. Suppose, a parallel arbiter has 5 bus arbiters. What will be the size of priority encoder and decoder respectively?

- (a) $4 \times 2, 2 \times 4$
- (b) $2 \times 4, 4 \times 2$
- (c) $3 \times 8, 8 \times 3$
- (d) $8 \times 3, 3 \times 8$

Ans. (d) : Arbiters $\rightarrow$ Arbiters are electronic devices that allocate access to shared resources.

Example $\rightarrow$

A bus arbiter is a device used a multi master bus system to decide which bus master will be allowed to control the bus for each bus cycle.

Since, it is given that the parallel arbiter has 5 bus arbiters. Hence, number of bits required to provide 5 bus arbiters are 3 bits (Since $2^3 = 8$).

Now encoder is a type of combinational circuit which consist of 2^n input lines and n output lines i.e $2^n \times n$ encoder. Now $n = 3$ bits. Therefore, size of encoder is 8×3 . Similarly decoder is a combinational circuit which consist of n-input lines and 2^n output lines, i.e, $n \times 2^n$ decoder. So size of decoder is 3×8 .

18. Consider the equation $(146)_b + (313)_{b-2} = (246)_8$ Which of the following is the value of b?

- (a) 8
- (b) 7
- (c) 10
- (d) 16

Ans. (b) : First convert each term is base 10 to find base and make equation

Method 1:-

First convert all into base 10

$$(146)_b + (313)_{b-2} = (246)_8$$

$$\Rightarrow b^2 + 4b + 6 + 3(b-2)^2 + 1(b-2) + 3 = 128 + 32 + 6$$

$$b^2 + 5b + 7 + 3(b^2 + 4 - 4b) = 166$$

$$b^2 + 5b + 7 + 3b^2 + 12 - 12b = 166$$

$$4b^2 - 7b + 19 - 166 = 0$$

$$4b^2 - 7b - 147 = 0$$

$$b = \frac{+7 \pm \sqrt{49 - 4 \times 4 \times -147}}{2 \times 4}$$

$$= \frac{7 \pm 49}{8} = \frac{56}{8} = 7$$

Method 2:- By elimination

1-put $b = 8$ convert each into base 1

$$(64 + 32 + 6)_{10} + (3 \times 36 + 6 \times 3)_{10} = (64 \times 2 + 32 + 6)_{10}$$

$$(219)_{10} \neq (166)_{10} \text{ false}$$

2-put $b = 7$
 $(49 + 28 + 6)_{10} + (75 + 5 + 3)_{10} = (64 \times 2 + 32 + 6)_{10}$
 $(166)_{10} = (166)_{10}$ True
 So 7 is right Answer.

19. How many address lines and data lines are required to provide a memory capacity of $16k \times 16$?

- (a) 10, 4 (b) 16, 16
 (c) 14, 16 (d) 4, 16

Ans. (c) : Memory capacity = $16k \times 16 = 2^{14} \times 2^4$
 Memory capacity is of the form = $2^m \times 2^n$
 Address lines required = $r = 14$
 Data lines required = $2^n = 16$

20. Suppose that a computer program takes 100 seconds of execution time on a computer with multiplication operation responsible for 80 seconds of this time. How much do you have to improve the speed of multiplication operation if you are asked to execute this program four times faster?

- (a) 14 times faster
 (b) 15 times faster
 (c) 16 times faster
 (d) 17 times faster

Ans. (c) : Total Execution time = time to execute multiplication operation + time for other operations.

Calculation $\rightarrow$ Total execution time = 100 seconds

Time to execute multiplication operation = 80 Seconds

Then time to execute other operation = $100 - 80$
 = 20 Seconds.

To execute the program four time faster, we have to improve the multiplication operation according to the question.

To execute the program four times faster.

Total execution time will become = $100/4 = 25$ Second

New time to perform multiplication operation

= new total execution time – time for other operations

New time to perform multiplication operation

= $25 - 20 = 5$ Sec.

So speed of multiplication operation is improved by = $80/5 = 16$ times

speed becomes 16 times fast.

21. Consider the following pseudo-code fragment in which an invariant for the loop is " $m \cdot x^k = p^n$ and $k \geq 0$ " (here, p and n are integer variables that have been initialized) :

/* Pre-conditions : $p \geq 1 \wedge n \geq 0$ */

/* Assume that overflow never occurs */

int $x = p$; int $K = n$; int $m = 1$;

while ($k < > 0$) {

if (k is odd) then $m = m \cdot x$;

$x = x \cdot x$;

$k = \lfloor k/2 \rfloor$; /*floor ($k/2$)*/

}

Which of the following must be true at the end of the while loop?

- (a) $x = p^n$
- (b) $m = p^n$
- (c) $p = x^n$
- (d) $p = m^n$

Ans. (b) : This pseudo code basically represents the exponentiation algorithm. With p as the base and n as the power and the final result is stored in m .

22. Consider the following C-code fragment running on a 32-bit x86 machine :

<pre>typedef struct { union { unsigned char a; unsigned short b; }U; unsigned char c; }S;</pre>	<pre>SB[10]; S*p = &B[4]; S*q = &B[5]; p → U.b = 0x1234; /*structure Stakes 32 – bits*/</pre>
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If M is the value of $q - p$ and N is the value of $((\text{int}) \& (p \rightarrow c)) - ((\text{int})P)$, then (M, N) is

- (a) (1, 1)
- (b) (3, 2)
- (c) (1, 2)
- (d) (4, 4)

Ans. (c) : In this, size of short int = 4 bytes
 $S * p = \& B [4];$ // p is having the address of $B [4]$
 $S * q = \& B [5];$ // q is having the address of $B [5]$
 suppose $B[4]$ address is 116 [take base address as 100].
 So $B [5]$ will come as 120
 $q - p$ will be = (address of $B[5]$ – address of $B[4]$)
 $q - p = (120 - 116)/4 = 4/4 = 1$
 Also M is equal to $q - p$ (given in the question)
 So $M = 1$
 Now, we have to find the value of $N =$
 $((\text{int}) \& (p \rightarrow c)) - ((\text{int}) p),$
 Here, c is a char and it is type casted to int $((\text{int}) \& (P \rightarrow C)) - ((\text{int}) p$ // address difference between p 's starting address and c address will be of 2 bytes only.
 So N becomes 2
 $\therefore (M, N) = (1, 2)$

23. What is the output of the following JAVA program?

```
public class Good {
    private int m;
    public Good (int m) {this m = m;}
    public Boolean equals (Good n) {return n.m = m;}
    public static void main (string args []) {
        Good m1 = new Good (22);
        Good m2 = new Good (22);
```

```

Object s1 = new Good (22);
Object s2 = new Good (22);
System . out . println (m1 equals (m2));
System . out . println (s1 equals (s2));
System . out . println (m1 equals (s2));
System . out . println (s1 equals (m2));
}
}

```

- (a) True, True, False, False
- (b) True, False, True, False
- (c) True, True, False, True
- (d) True, False, False, False

Ans. (d) : System out println (m1.equals (m2)),	True
System . out . println (s ₁ equals (s ₂));	False
System . out . println (m ₁ equals (s ₂));	False
System . out . println (s ₁ equals (m ₂));	False

24. Consider the following C++ function f() :
unsigned int f (unsigned int n){

```

    Unsigned int b = 0
    while (n) {
    b += n & 1;
    n >> 1;
    }
    return b;
}

```

The function f() returns the int that represents the __p__ in the binary representation of positive integer n, where P is

- (a) number of 0' s
- (b) number of bits
- (c) number of consecutive 1's
- (d) number of 1's

Ans. (d) : b += n & 1; binary AND operation and then add result with b consider n as 5; So binary representation of n is 101 initially b = 0 B += n & 1; // It means b = 0 + 101 & 001; b = 0 + convert to decimal (001); b = 1 n >> // n is right shifted to 1 so, n becomes 010 b += n & 1; this gives b = 1 + 010 & 001; b = 1 + 0; b = 1 Again, n is right shifted; n becomes 001 b += n & 1; // b = 1 + 001 & 001; b = 1 + 1 = 2 Again, n is right shifted n becomes 000 while (n) //It becomes false program will return the final value of b; which is 2. Which is equal to the number of 1's in the binary representation of n.

25. Which of the following statements is/are true?

P : In a scripting language like JavaScript, types are typically associated with values, not variables.

Q : It is not possible to show images on a web page without the tag of HTML.

Select the correct answer from the options given below :

- (a) P only
- (b) Q only
- (c) Both P and Q
- (d) Neither P nor Q

Ans. (a) : P → In a scripting language like Java script, types are typically associated with values, not variables. Java script is a client side as well as server-side scripting language. Java Script data types are : number, string, Boolean, object undefined. Basically, the type of a value determines the operations and methods performed on it. Variables are not directly associated with any particular type; any variable can be assigned values of all type.

Images can be shown on a web page without the use of image tag of HTML. We can use div by giving the URL of the image in that.

26. Which of the following statements is/are true?

P : An XML document with correct syntax as specified by W3C is called "Well Formed".

Q : An XML document validated against a DTD is both "Well formed" and "Valid".

R : <xml version = "1 . 0" encoding = "UTF – 8"> is syntactically correct declaration for the version of an XML document.

Select the correct answer from the options given below :

- (a) P and Q only
- (b) P and R only
- (c) Q and R only
- (d) All of P, Q and R

Ans. (a) : XML - XML stands for extensible markup language. It is a meta markup language for text documents/data. It allows to define language to represent text. It is designed to describe data not to display data. XML tags are not predefined.

Some points in XML are as follows :

* An XML document is well formed if it satisfies the rule specified by the W 3C.

* A well - formed XML has a proper starting and closing tag.

* An XML document validated against a DTD is both "Well formed" and "valid"

* Purpose of DTD is to define the structure and legal elements and attributes of a XML document.

* XML does not require DTD.

* XML carries the data; it does not represent it.

* XML can be used for off loading and reloading of database.

XML declaration syntax is :

<?Xml version = "1.0"

encoding = "UTF – 8"?

27. Consider a raster system with resolution 640 by 480. What size is frame buffer (in bytes) for this system to store 12 bits per pixel?

- (a) 450 kilobytes
- (b) 500 kilobytes
- (c) 350 kilobytes
- (d) 400 kilobytes

Ans. (a) : Raster scan system : In this electron beam move on the screen from top to bottom covering each row one by one.

Explanation :

Given resolution of the system is : 640×480 we have to store 12 bits per pixel.

Frame buffer size = resolution (number of pixels)
x number of bits per pixel

Frame buffer size = $(640 \times 480) \times 12$
= 3686400 bits

convert it into bytes.

frame buffer size = $3686400/8 = 460800$ Byte
= 450 KB

28. Consider the following statement regarding 2D transforms in computer graphics :

S1 : $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a 2×2 matrix that reflects (mirrors)

only 2D point about the X-axis.

S2 : A 2×2 matrix which mirror any 2D point about the X-axis, is a rotation matrix.

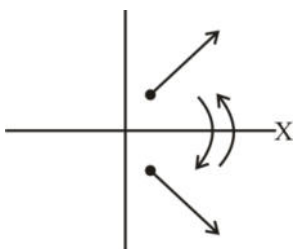
What can you say about the statements S1 and S2?

- (a) Both S1 and S2 are true
- (b) Only S1 is true
- (c) Only S2 is true
- (d) Both S1 and S2 are false

Ans. (b) : First matrix is reflection matrix about x-axis
 S_1 is true always.

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is trans formation matrix for reflection about x - axis.

S_2 is false because reflection about x - axis will give rotation of $\theta = 90^\circ$



So if clockwise rotation occurs

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and if anticlockwise rotation occurs-

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Both of these matrix are not same as matrix given in S_1 .
So S_2 is false.

29. In the context of 3D computer graphics, which of the following statements is/are true?

P : Orthographic transformations keep parallel lines parallel.

Q : Orthographic transformations are affine transformations.

Select the correct answer from the options given below:

- (a) Both P and Q
- (b) Neither P nor Q
- (c) Only P
- (d) Only Q

Ans. (a) : orthogonal projections are parallel projection
Each line that is originally parallel will be parallel after this transformation-

orthographic projection (sometimes referred to as orthogonal projection, used to be called analemma is a means of representing three dimensional objects in two dimensions. It is a form of parallel projection, in which all the projection lines are orthogonal which to the projection plane.

In Euclidean geometry, an affine transformation or an affinity (from the Latin, affini's, connected with is a geometric transformation that preserve lines and parallelism (but not necessarily distance and angles).

The orthographic projection can be represent by a affine transformation.

30. Using the phong reflectance model, the strength of the specular highlight is determined by the angle between

- (a) the view vector and the normal vector
- (b) the light vector and the normal vector
- (c) the light vector and the reflected vector
- (d) the reflected vector and the view vector

Ans. (d) : Phong reflectance model :

* It is local illumination model. In real world, light is shown with the help of its spectrum. In computer graphics, light color is made of a triplet that is of three colours RGB(red, green and blue).

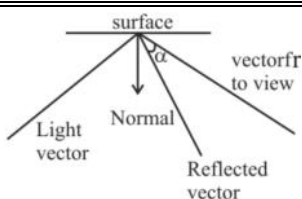
* The reflectance spectrum is a triple of percent the light seen on a surface can be divided into four components such as : emissive component, ambient component, diffusive component, specula component.

Specular component :

This component shows the reflected light in a specific direction. It reflects on the basis of reflection of incident light. The specular light depends on the position of observer with respect to the surface.

Specular component finds the strength which is the multiplication of specular surface constant with specular intensity and with dot product of reflection direction and the direction of surface towards the view. It means it is determined by the angle between reflected and view vector.

So ans. is d. the reflected vector & the view vector.



31. Which of the following has same expressive power with regard to relational query language?

- (A) Relational algebra and domain relational calculus
 - (B) Relational algebra and tuples relational calculus
 - (C) Relational algebra and domain relational calculus restricted to safe expression
 - (D) Relational algebra and tuples relational calculus restricted to safe expression
- (a) (A) and (B) only
 (b) (C) and (D) only
 (c) (A) and (C) only
 (d) (B) and (D) only

Ans. (b) :

* Relational algebra is a procedural query language, which takes instances of relations as input and yields instances of relations as output. It uses operation (selection, projection, union etc) to perform queries

* Relational calculus consists of two calculi, the tuple relational calculus and the domain relational calculus, that are part of the relational model for databases and provide a declarative way to specify database queries.

* Relational algebra, Domain relational calculus restricted to safe expression and tuple relational calculus restricted to safe expression have the same expressive power.

Relation algebra's expressive power is not equal to Domain relational calculus or Tuple relational calculus unless they are restricted to safe operation.

32. Which of the following features is supported in the relational database model?

- (a) Complex data-types
- (b) Multivalued attributes
- (c) Associations with multiplicities
- (d) Generalization relationships

Ans. (c) : RDBMS (Relational database management system) is structured in database tables. It stores the data into collection of tables which are related by some fields. Goal of RDBMS is to generate a set of relation scheme that allows as to store information without unnecessary redundancy and to retrieve information easily.

Composite data types : Those data types that are composed of other existing data types such as date, images, time etc. They are used as objects. Complex data types are not allowed in RDBMS. These are stored in LOB, they cannot be accessed directly in RDBMS.

Multivalued attributed : An attribute that contains more than one value for with it. It is not allowed in RDBMS. An attribute can contain only one value at one time.

Association with multiplicities : → Multiplicity means the cardinality of a particular entity which can be associated with other entities. In terms of data base, it means participation and cardinality of the relation. It is allowed in RDBMS.

Generalization Relationship : Generalization means combination of lower entities to form a higher entity. Entities are combined which have common attribute.

33. With respect to relational algebra, which of the following operations are included from mathematical set theory?

(A) Join

(B) Intersection

(C) Cartesian product

(D) Project

(a) (A) and (D)

(b) (B) and (C)

(c) (C) and (D)

(d) (B) and (D)

Ans. (b) : Relational Algebra :→ It is a procedure oriented language. In this, we use relation as values instead of numbers. It works on relational model. Main work is to retrieve data, perform operations such as insert, delete, update etc.

* Basic operation in relational algebra are select, Project, union, set difference, Cartesian product, rename.

* other operations are natural join, intersection division.

* from the above options, operator that are included from mathematical set theory are intersection and Cartesian product.

* In mathematical set theory, intersection is used to find the common elements between two sets. It works same in relational algebra.

* Cartesian product is used to find the multiplication of two sets by including common element only once.

- 34. In relational databases, if relation R is in BCNF, then which of the following is true about relation R?**
- (a) R is in 4NF
 - (b) R is not in 1 NF
 - (c) R is in 2NF and not in 3NF
 - (d) R is in 2NF and 3NF

Ans. (d) : relation is in 'N' Normal form it means it must be in 1NF, 2NF, N-1 NF.
 standard Normal form hierarchy :→
 1NF → 2NF → 3NF → BCNF
 So Relation is in BCNF then it must be in 1NF, 2NF, & 3NF Hence option 4 is correct.

- 35. Which of the following key constraints is required for functioning of foreign key in the context of relational data bases?**
- (a) Unique key
 - (b) Primary key
 - (c) Candidate key
 - (d) Check key

Ans. (b) :

- * **Candidate Key** →
 - It can be null,
 - It is null in that such case
 - Foreign key point to primary key.
- * **Alternate Key** → unique and null
- * If candidate key is null in such case foreign key points to primary key.
 though foreign keys can always point to primary key but not to candidate key.

- 36. In relational database management, which of the following is/are property/properties of candidate key?**
- P : Uniqueness**
Q : Irreducibility
- (a) P only
 - (b) Q only
 - (c) Both P and Q
 - (d) Neither P nor Q

Ans. (c) : candidate key : It uniquely identifies the tuples in a relation. A candidate key is a super key with no redundant attribute. In a relation, there can be more than one candidate key.

As, candidate uniquely identifies the tuples in a relation. So, first property of the candidate key is its uniqueness. second, candidate key does not contain redundant attribute. We cannot further reduce a candidate key. If a candidate key is reduced, then it can cause partial dependency. second property is its irreducibility.

Example :

A relation with four attributes : R(A,B,C,D).

functional dependencies are :

$A \rightarrow C$

$B \rightarrow D$

In this, AB is the candidate key.

AB uniquely tuples in relation R also AB cannot be reduced further.

37. Which of the following statements are DML statements?

- (A) Update [tablename]
Set [columnname] = VALUE
- (B) Delete [tablename]
- (C) Select * from [tablename]
 - (a) (A) and (B)
 - (b) (A) and (D)
 - (c) (A), (B), and (C)
 - (d) (B) and (C)

Ans. (c) : A data manipulation language (DML) is a language that enables users to access or manipulate data as organized by the appropriate model.

Example of DML commands : select, insert, delete, update etc.

therefore a, b and c are DML statements.

A database system provides a **data-definition language (DDL)** to specify the database schema example of DDL commands : Create, drop, alter, rename, truncate etc.

A **data control language (DCL)** is used to control access (Authorization) to data stored in a database example of DCL commands : grant and revoke.

38. Hadoop (a big data tool) works with number of related tools, Choose from the following, the common tools included into Hadoop :

- (a) MySQL, Google API and Map reduce
- (b) Map reduce, Scala and Hummer
- (c) Map reduce, H Base and Hive
- (d) Map reduce, Hummer and Heron

Ans. (c) : Hadoop is an open source platform providing highly reliable, scalable, distributed processing of large data sets using simple programming models.

Hadoop provides a reliable shared storage and analysis system. In this, storage, is provide by HDFS (Hadoop distributed file system) and analysis by map reduce.

HDFS breaks a file into chunks and distributed them across the nodes of the cluster. It stores large amount of data on local disk across a distributed cluster of computers. It is written in Java. Hadoop work with the tools : Map Reduce, H base, hive.

Map Reduce A distributed data processing model and execution environment that runs on large clusters of commodity machines.

Hbase → It is a distributed, column-oriented database. HBase uses HDFS for its underlying storage and supports both batch style computations using map reduce and point queries.

HIVE → It is a distributed data ware house. Hive manages data stored in HDFS and provides a query language based on SQL.

39. Following table has two attributes Employee id and Manager id, where Employee is a primary key and manager id is a foreign key referencing Employee id with on delete cascade:

| Employee_id | Manager_id |
|-------------|------------|
| 20 | 40 |
| 25 | 40 |
| 30 | 35 |
| 35 | 20 |
| 40 | 45 |
| 45 | 25 |

On deleting the table (20, 40), the set of other tuples that must be deleted to maintain the referential integrity of table is

- (a) (30, 35) only
- (b) (30, 35) and (35, 20) only
- (c) (35, 20) only
- (d) (40, 45) and (25, 40) only

Ans. (b) : Manager-id is foriegn key referring to employee-id (primary key) of the same table

| Employee_id | Manager_id |
|----------------|----------------|
| 20 (Deleted 1) | 40 (Deleted 1) |
| 25 | 40 |
| 30 (Deleted 3) | 35 (Deleted 3) |
| 35 (Deleted 3) | 20 (Deleted 2) |
| 40 | 45 |
| 45 | 25 |

(20, 40) is deleted, that is employee-id = 20 is deleted therefore manager-id = 20 should be deleted to maintain the referential integrity of table.

If Manager-id = 20 row is deleted, then employee-id = 35 is also deleted

therefore manager-id = 35 should also be deleted to maintain the referential integrity of table.

If manager-id = 35 row is deleted then employee-id = 30 is also deleted since If manager-id $\neq$ 35 we stop here.

Table after deletion

| | Employee_id | Manager_id |
|--|-------------|------------|
| | 25 | 40 |
| | 40 | 45 |
| | 45 | 25 |

Apart from (20, 40) (30, 20) also gets deleted to maintain the referential integrity of table.

40. K-mean clustering algorithm has clustered the given 8 observations into 3 clusters after 1st iteration as follows:

C1 : {(3, 3), (5, 5), (7, 7)}

C2 : {(0, 6), (6, 0), (3, 0)}

C3 : {(8, 8), (4, 4)}

What will be the Manhattan distance for observation (4, 4) from cluster centroid C1 in second iteration?

- (a) 2
- (b) $\sqrt{2}$
- (c) 0
- (d) 18

Ans. (a) : Manhattan distance is the distance between two points measured at right angle along both the X-axis and Y-axis.

formula- It is calculated as : if there is a point P(X_1 , Y_1) and Q(X_2 , Y_2). Then Manhattan distance is

$$|X_2 - X_1| + |Y_2 - Y_1|$$

calculation :

C1 : {(3, 3), (5, 5), (7, 7)}

C2 : {(0, 6), (6, 0), (3, 0)}

C3 : {(8, 8), (4, 4)}

first point is (4, 4)

we have to find the Manhattan distance for cluster C1 in 2nd iteration :

for 2nd iteration : points will be

$$| \left(\frac{1}{3} \left(3 + 5 + 7 \right) \right) - 4 | + | \left(\frac{1}{3} \left(3 + 5 + 7 \right) \right) - 4 | = | 5 - 4 | + | 5 - 4 |$$

$$\text{So Manhattan distance} = |5 - 4| + |5 - 4| = 1 + 1 = 2$$

41. Consider a disk system with 100 cylinders. The requests to access the cylinders occur in the following sequence :

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1ms to move from the cylinder to adjacent one and the shortest seek time first policy is used?

- (a) 357 ms
- (b) 238 ms
- (c) 276 ms
- (d) 119 ms

Ans. (d) :

$$\begin{aligned}
 &= (50 - 34) + (34 - 20) + (20 - 19) + (19 - 15) + (15 - 10) + (10 - 7) + (7 - 6) + (6 - 4) + (4 - 2) + (73 - 2) \\
 &= 16 + 14 + 1 + 4 + 5 + 3 + 1 + 2 + 2 + 71 \\
 &= 119 \text{ ms}
 \end{aligned}$$

42. Match List-I with List-II

| | List-I | | List-II |
|-----|-----------|-------|-----------------------|
| (A) | Disk | (i) | Thread |
| (B) | CPU | (ii) | Signal |
| (C) | Memory | (iii) | File system |
| (D) | Interrupt | (iv) | Virtual address space |

Choose the correct option from those given below :

- (a) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
- (b) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)
- (c) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)
- (d) (A)-(ii), (B)-(iv), (C)-(iii), (D)-(i)

Ans. (b) :

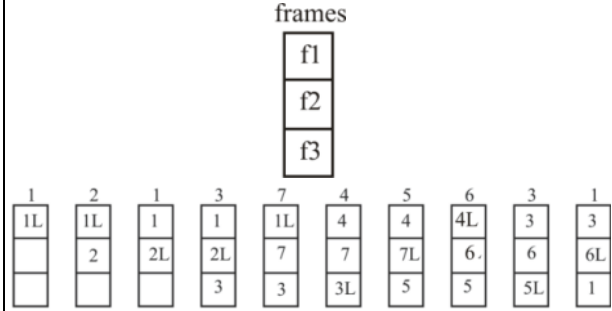
| | List-I | | List-II |
|-----|-----------|-------|-----------------------|
| (A) | Disk | (iii) | File system |
| (B) | CPU | (i) | Thread |
| (C) | Memory | (iv) | Virtual address space |
| (D) | Interrupt | (ii) | Signal |

43. Consider that a process has been allocated 3 frames and has a sequence of page referencing as 1, 2, 1, 3, 7, 4, 5, 6, 3, 1.

What shall be the difference in page faults for the above string using the algorithms of LRU and optimal page replacement for referencing the string?

- (a) 2
- (b) 0
- (c) 1
- (d) 3

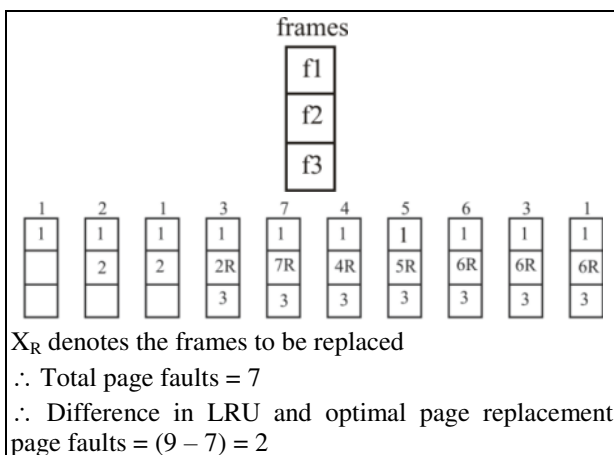
Ans. (a) : LRU ALGORITHM :- Replace the least recently used page if frames are not empty.



X_L denotes the frames to be replaced

$\therefore$ Total page faults = 9

Optimal page Replacement Algorithm :- Replace the page which will not be used for longest duration of time if the frames are not empty.



44. Which of the following are Not shared by the threads of the same process?

- (A) Stack
 - (B) Registers
 - (C) Address space
 - (D) Message queue
- (a) (A) and (D)
- (b) (B) and (C)
- (c) (A) and (B)
- (d) (A), (B) and (C)

Ans. (c) : "In general each thread has its own registers (including its own program counter), its own stack pointer, and its own stack. Everything else is shared between the threads sharing a process, why register cannot be shared?

Each thread needs its own program counter value (which is a register)

why stack cannot be shared?

Each thread may have its own calling sequence.

So stack and registers is correct.

45. Consider three CPU intensive processes, which require 10, 20 and 30 units of time and arrive at times 0, 2 and 6 respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and the end.

- (a) 4
- (b) 2
- (c) 3
- (d) 1

Ans. (b) : Let three process be P0, P1 and P2 with arrival times 0, 2 and 6 respectively and CPU burst times 10, 20 and 30 respectively. At time 0, P0 is the only available process so it runs. At time 2, P1 arrives, but P0 has the shortest remaining time, So it continues. At time 6, P2 arrives, but P0 has the shortest remaining time process. At time 30, P2 is scheduled. Only two context switches are needed. P0 to P1 and P1 to P2. Processes executes as per the following Gantt chart

| | | |
|----|----|----|
| P1 | P2 | P3 |
| 0 | 10 | 30 |
| | | 60 |

So, here only 2 switching possible (when we did not consider the starting and ending switching)

Now here might be confusion that at $t = 2$. P1 is preempted and check that available process have shortest job time or not, but he did not get anyone So it should not be consider as context switching.

46. At a particular time of computation, the value of a counting semaphore is 7. Then 20 P (wait) operations and 15V (signal) operations are completed on this semaphore. What is the resulting value of the semaphore?

- (a) 28
- (b) 12
- (c) 2
- (d) 42

Ans. (c) : $I \rightarrow P$ represent wait and v represent signal. P operation will decrease the value by 1 every time and Y operation will increase the value by 1 every time.

Value of counting semaphore = 7

After 20 P operations value of semaphore = $7 - 20$
 $= -13$

After 15 v operations value of semaphore
 $= -13 + 15 = 2$

47. The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

- (a) page size
- (b) physical size of memory
- (c) the instruction set architecture
- (d) number of processes in memory

Ans. (c) : There are two important task in virtual memory management : a page replacement strategy and a frame-allocation strategy. Frame allocation strategy say gives the idea of minimum number of frames that a process must be allocated is dependent on system architecture, and corresponds to the number of pages that could be touched by a single (machine) instruction.

So the instruction set architecture is right answer.

48. A computer has six tape drives with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

- (a) 5
- (b) 4
- (c) 3
- (d) 6

Ans. (a) : Given tape drive = 6 and each process may need 2 drive.

When we give 1 drive to 1 process then total process will be 6 but in this case there will definitely deadlock occur because every process contain 1 drive and waiting for another drive which is hold by other process

therefore when we reduce 1 process then system to be deadlock free.

Hence maximum value of $n = 6 - 1 = 5$

49. A processor can support a maximum memory of 4 GB where memory is word addressable and a word is 2 bytes. What will be the size of the address bus of the processor?

- (a) At least 28 bits
- (b) At least 2 bytes
- (c) At least 31 bits
- (d) Minimum 4 bytes

Ans. (c) : Maximum Memory = 4GB = 2^{32} bytes
size of a word = 2 bytes

Therefore, Number of words = $2^{32}/2 = 2^{31}$

So, we require 31 bits for the address bus of the processor.

50. Which of the following UNIX/Linux pipes will count the number of lines in all the file having c and h as their extension in the current working directory?

- (a) `cat*ch| wc-1`
- (b) `cat*[c-h]|wc-1`
- (c) `cat*[ch]|ls-1`
- (d) `cat*[ch]|wc-1`

Ans. (d) : Pipes in Linux/Unix : Pipe is a command which produces an output which serves as input for next command. It works as a pipeline structure in computer.

Explanation :

We have to find the number of lines in all the files having .c and .h as their extension in the current working directory.

WC- in Unix/Linux, "WC" is used for word count.

Cat- it is used to display the content of the files.

Option 4- `Cat * .[ch] //it represent the content of files having .c and .h extension`

Option 2 →

`Cat * [c -h]/ WC-L //it also includes the files with c and .h extension but it also considers the other extensions that come between because it is a range which is not required in this question.`

So `cat * [ch]/WC-L`

It will count the number of Lines in all the files having .c and .h as their extension in current working directory.

51. Which of the following statements is/are true?

P : In software engineering, defects that are discovered earlier are more expensive to fix.

Q : A software design is said to be a good design, if the components are strongly cohesive and weakly coupled.

Select the correct answer from the options given below :

- (a) P only
- (b) Q only

- (c) P and Q
- (d) Neither P nor Q

Ans. (b) : statements P : In software engineering, defects that are discovered earlier are more expensive to fix.

This statement is not correct. In software engineering, defects which are early detected are easier to fix. It reduces error fixing cost. It increases the productivity. Early defect detection improves the overall quality of software.

Statement Q : A software design is said to be a good design, if the components are strongly cohesive and weakly coupled.

Coupling : Coupling in software engineering mean interdependence between the modules. Cohesion means how the elements of one module are related with each other. It means intra-dependency between the module. For a good software design, low coupling and high cohesion is desired.

52. The M components in MVC are responsible for

- (a) user interface
- (b) security of the system
- (c) business logic and domain objects
- (d) translating between user interface actions/events and operations on the domain objects

Ans. (c) : MODEL-VIEW-CONTROLLER (MVC):

- * MVC refers to three different realms of responsibility in the design of a graphical user interface.
 - * when the MVC pattern is applied to a component the model consists of the data that represents the current state of the component.
 - * The view is simply the visual presentation of the component on the screen.
 - * The controller is the aspect of the component that carries out actions in response to events generated by the user (or by other source such as timers).
- The idea is to assign responsibility for the model, the view, and the controller to different objects.

53. In the context of software testing, which of the following statements is/are Not correct?

P : A minimal test set that achieves 100% path coverage will also achieve 100% statement coverage.

Q : A minimal test set that achieves 100% path coverage will generally detect more faults than one that achieves 100% statement coverage.

R : A minimal test that achieves 100% statement coverage will generally detect more faults than one that achieves 100% branch coverage.

- (a) R only
- (b) Q only
- (c) P and Q only
- (d) Q and R only

Ans. (a) : A minimal test set that achieves 100% path coverage will also achieve 100% statement coverage. This statement is correct statement coverage means executing each statement at least once. Where as Path coverage is in which every distinct entry- exit path through the code is executed at least once in some test. It is a white box technique. It tells about how well the source code has been made. A minimal test set that achieves 100% path coverage will also achieve 100% statement coverage.

Statement Q : A minimal test set that achieves 100% path coverage will generally detect more faults than one that achieves 100% statement coverage Sometimes, during minimal test it can be the possibility that test suite produce 100% statement coverage will miss certain defects related to control structure. So this statement is correct.

Statements R : A minimal test set that achieves 100% statement coverage, will generally detect more faults than one that achieves 100% branch coverage.

Branch coverage means, for each branch is chosen at least once in some test. It includes statement coverage. As, in case of statement coverage, test suite sometimes cannot detect all faults. So this statement is not correct that minimal test set that achieves 100% statement coverage will generally detect more faults than one that achieve 100% branch coverage.

54. Software reuse is

- (a) the process of analysing software with the objective of recovering its design and specification
- (b) the process of using existing software artifacts and knowledge to build new software
- (c) concerned with reimplementing legacy system to make them more maintainable
- (d) the process of analysing software to create a representation of higher level of a abstraction and breaking software down into its parts to see how it works

Ans. (b) : "Software reuse is the process of creating new software system from predefined software components."

55. Which of the following terms best describes Git?

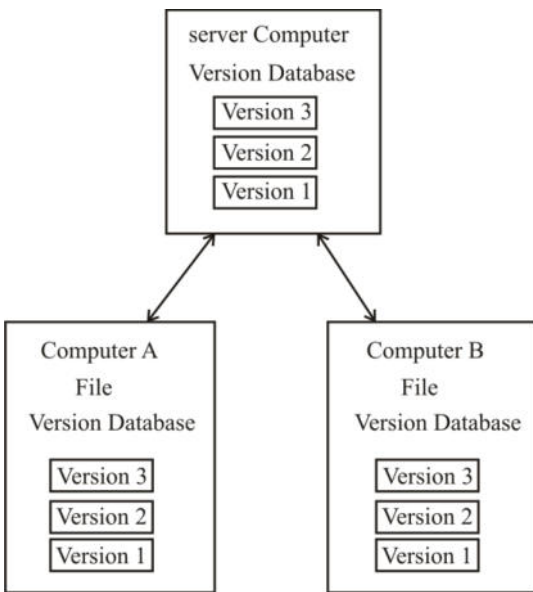
- (a) Issue Tracking System
- (b) Integrated development Environment
- (c) Distributed Version Control System
- (d) Web-based Repository Hosting Service

Ans. (c) : Git is an open source distributed version control system. Version control is a system that records changes to a file or set of files over time. So that you can recall specific versions later.

- * Local version control system : In this, files are copied into another directory. This approach is very common. In this it is easy to forget which directory you are in and accidentally write to the wrong file.
- * Centralized version control system : The next issue people encounter that they need to collaborate with

developers on other systems. These systems have a single server that contains all versioned files and number of clients that check out file from central place. Main disadvantage is that a single point of failure can cause the entire system.

* Distributed version control system : In this, client do not just check out the latest snapshot of the files, they fully mirror the repository. Thus, if any server dies and these systems were collaborating via it, any of the client repositories can be copied back up to the server to restore it. Every clone is fully backup of all the data.



56. A Web application and its support environment has not been fully fortified against attack. Web engineers estimate that the likelihood of repelling an attack is only 30 percent. The application does not contain sensitive or controversial information, so the threat probability is 25 percent. What is the integrity of the web application?

- (a) 0.625
- (b) 0.725
- (c) 0.775
- (d) 0.825

Ans. (d) : Integrity of web application :
 Integrity of web application can be found out using the security and threat in that.
 Formula :

$$\text{Integrity} = [1 - \text{threat} (1 - \text{security})]$$
 Calculation -
 It is given that, probability or Likelihood of repelling an attack = 30%
 i.e probability of security = 0.30
 Threat probability is given as 25% or 0.25
 So integrity = $[1 - 0.25 (1 - 0.30)]$

$$\text{integrity} = [1 - 0.25 (0.70)]$$

$$\text{Integrity} = 1 - 0.175 = 0.825$$

57. Match List-I with List-II

| | List-I (Software Process Models) | | List-II (Software Systems) |
|-----|---|-------|---|
| (A) | Waterfall model | (i) | e-business software that starts with only the basic functionalities and then moves to more advanced features |
| (B) | Incremental development | (ii) | An inventory control system for a super-market to be developed within three months |
| (C) | Prototyping | (iii) | A virtual reality system for simulating vehicle navigation in a highway |
| (D) | RAD | (iv) | Automate the manual system for student record maintenance in a school |

Choose the correct option from those given below :

- (a) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
- (b) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)
- (c) (A)-(iii), (B)-(ii), (C)-(iv), (D)-(i)
- (d) (A)-(iv), (B)-(i), (C)-(iii), (D)-(ii)

Ans. (d) : (A) Waterfall model- (iv) Automate the manual system for student record maintenance in a school

(B) Incremental development - (i) e-business software that starts with only the basic functionalities and then moves to more advanced features

(C) Prototyping - (iii) A virtual reality system for simulating vehicle navigation in a highway

(D) RAD - (ii) An inventory control system for a super-market to be developed within three months

58. Software products need adaptive maintenance for which of the following reasons?

- (a) To rectify bugs observed while the system is in use
- (b) When the customers need the product to run on new platforms

- (c) To support the new features that users want it to support
- (d) To overcome wear and tear caused by the repeated use of the software

Ans. (b) : Maintenance :- Maintenance can be referred as a process of enhancement in the software product according to the changing requirements of users.

Four types of maintenance :

- 1– Adaptive : Modifying the system to cope with changes in the software environment (DBMS, OS),
- 2– Perfective : implementing new or changed user requirements which concern functional enhancements to the software.
- 3– Corrective : diagnosing and fixing errors, possibly ones found by users.
- 4– Preventive : Increasing software maintainability or reliability to prevent problems in the future.

59. Which of the following are the primary objectives of risk monitoring in software project tracking?

P : To assess whether predicted risks do, in fact, occur

Q : To ensure that risk aversion steps defined for the risk are being properly applied

R : To collect information that can be used for future risk analysis

- (a) Only P and Q
- (b) Only P and R
- (c) Only Q and R
- (d) All of P,Q,R

Ans. (d) : Risk monitoring and control in software engineering ensures the execution of risk plans and help in reducing the risk in software projects.

It keeps track of identifies risks. It monitors the already occurring risks and can identify new risks in project execution.

Primary objectives of risk monitoring are :

- 1) To assess whether predicted risk do occur.
- 2) To ensure that risk aversion steps defined for the risk are being properly applied.
- 3) To collect information that can be used for future risk analysis.
- 4) To verify that risk actions are as effective as expected.
- 5) Proper procedures are followed or not.
- 6) To identify the new risk that were not occurred previously.
- 7) To identify whether the risk has changed or declined.

Risk monitoring helps in identifying the events that has a serious effect in controlling the risks It can produce alternate path for execution of events. It provides a good quality software after monitoring and controlling all the risks that can occur during the execution of software project.

60. Software validation mainly checks for inconsistencies between

- (a) use cases and user requirements
- (b) implementation and system design blueprints
- (c) detailed specifications and user requirements
- (d) functional specifications and use cases

Ans. (c) : Software validation means "are we building the right product".

Validation means that product meets the customers expectations. It goes beyond checking it meets its specification as system specifications don't always accurately reflect the real need of users.

Validation refers to the process of evaluating software at the end of its development to ensure that it is free from failure and implies with its requirement A failure is defined as incorrect product behaviors often the validation occurs through the utilization of various testing approaches.

There are two ways to perform software validation internal and external.

1– Internal validation assumes that goals of stakeholding were correctly understood.

2– External validation is performed by asking the stake holders if the software meets their needs.

61. Match List-I with List-II

| | List-I | | List-II |
|----|--------------------------------|-------|-----------------|
| A. | Prim's algorithm | (i) | $O(V^3 \log V)$ |
| B. | Dijkstra's algorithm | (ii) | $O(VE^2)$ |
| C. | Faster all-pairs shortest path | (iii) | $O(E \log V)$ |
| D. | Edmonds-Karp algorithm | (iv) | $O(V^2)$ |

Choose the correct option from those given below :

- (a) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
- (b) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
- (c) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)
- (d) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)

Ans. (b) : Prim's Algorithm :- It is used to find the minimum spanning tree for a weighted undirected graph. It uses greedy approach to find the MST for a graph. Unlike the Kruskal's algorithm, we add vertex to growing spanning tree in this. It there are v vertices and E edges in the graph. Then time complexity to for MST is $O(E \log V)$

Dijkstra's Algorithm :-

It is used to find the shortest path from source vertex to all other vertices in a graph. Initially, distance from source to source is considered as 0 and to all other vertices is infinity. It uses the condition "current vertex distance edge weight < next vertex distance" to find shortest distance. Time complexity for this is $O(v^2)$

Faster all pair shortest path : It finds shortest the distance from a vertex to all other vertices in a graph. It

there are E edges and V vertices in a graph, then it takes $O(v^2 \log v)$ time to find the shortest path.

Edmonds-Karp algorithm :- It is used to find the maximum flow in a network. It uses BFS in ford fulkerson method. Its time complexity is $O(V.E^2)$

62. There are many sorting algorithms based on comparison. The running time of heapsort algorithm is $O(N \lg n)$. Like P, but unlike Q, heapsort sorts in place where (P, Q) is equal to

- (a) Merge sort, Quick sort
- (b) Quick, sort, insertion sort
- (c) Insertion sort, Quick sort
- (d) Insertion sort, Merge sort

Ans. (d) : The question asks specifically for in-place sorting algorithms, which means the algorithms in which only $O(1)$ space is used.

Insertion sort :

In insertion sort, the worst case takes $O(n^2)$ time, worst case of insertion sort is when elements are sorted in reverse order. It is an in-place sorting algorithm.

In that case the number of comparisons will be like.

Merge Sort –

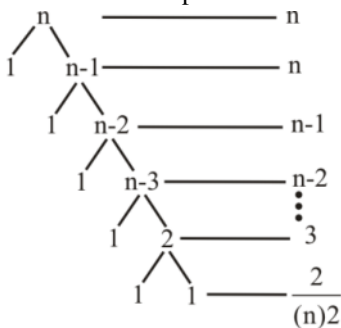
In merge sort, the worst case takes $O(n \log n)$ time. Merge sort is based on the divide and conquer approach. It is an out-of-place sorting algorithm. Recurrence relation for merge sort will become

$$T(n) = 2T(n/2) + O(n)$$

$$T(n) = n + O(n)$$

$$T(n) = n \times \log n$$

Quicksort : In Quicksort, the worst case takes $O(n^2)$ time. The worst case of quicksort is when the first or the last element is chosen as the pivot element.



Recurrence relation for quick sort $T(n) = T(n-1) + (n)$

This will give the worst case time Complexity as $O(n^2)$.

It is an in-place sorting algorithm.

Conclusion : P needs to be an in-place sorting algorithm, and Q needs to be an out-of-place sort in algorithm. Hence option d is correct

63. Consider the Euler's phi function given by

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

where p runs over all the primes dividing n . What is the value of $\phi(45)$?

- (a) 3
- (b) 12
- (c) 6
- (d) 24

Ans. (d) : If P is prime, then $\phi(p) = (p - 1)$

If P is a prime and it's power is k then

$$\phi(\text{p}^{\text{k}}) = \text{p}^{\text{k}} \left(1 - \frac{1}{\text{p}}\right)$$

Calculation →

$$\phi(45) = \phi(9 \times 5)$$

$$= \phi(3^2) \times \phi(5)$$

$$= (3^2 - 3) \times (5 - 1)$$

$$= 6 \times 4 = 24$$

the value of $\phi(45)$ is 24

64. Which of the following is best running time to sort n integers in the range 0 to $n^2 - 1$?

- (a) $O(\lg n)$
- (b) $O(n)$
- (c) $O(n \lg n)$
- (d) $O(N^2)$

Ans. (b) : Sort the integers in the range from 0 to $n^2 - 1$

If we take any comparison - based sorting algorithms for this, it will take $O(n \log n)$ time.

Radix sort is used for sorting this range integers. Radix sort takes $O(n)$ time for this.

Algorithm :

For any digit in the integer, varies from least significant digit to the most significant digit of the number. Sort the input array using count sort algorithm according to the digit.

Note : Radix sort depends on digits. It is more flexible than comparison based sorting algorithm. It solve the given problem in Linear time.

65. Which of the following is application of depth-first search?

- (a) Only topological sort
- (b) Only strongly connected components
- (c) Both topological sort and strongly connected components
- (d) Neither topological sort nor strongly connected components

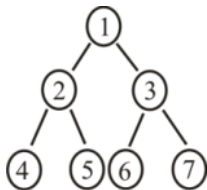
Ans. (c) : Depth first search: Depth first search algorithm is used to traverse the vertices of a graph. It uses the concept of back tracking.

Depth first search uses a data structure for its implementation. Which is known as stack.

Application of depth first search :

- 1) It is used to find the strongly connected components.
- 2) It helps in topological sorting
- 3) To check the bipartiteness of a graph.

- 4) It helps in detecting cycles in a graph.
 5) To find articulation points in a graph.



DFS traversal for this – 1, 2, 4, 5, 3, 6, 7

66. Consider double hashing of the form
 $h(K, i) = (h_1(k) + ih_2(k) \bmod m)$
 where $h_1(k) = k \bmod m$
 $h_2(k) = 1 + (K \bmod n)$
 where $n = m - 1$ and $m = 701$
 For $k = 123456$, what is the difference between first and second probes in terms of slots?
 (a) 255
 (b) 256
 (c) 257
 (d) 258

Ans. (c) : Data :

$$h(k, i) = (h_1(k) + ih_2(k)) \bmod m$$

$$\text{where } h_1(k) = k \bmod m, h_2(k) = 1 + (k \bmod n)$$

$$n = m - 1, m = 701 \text{ and } k = 123456$$

Calculation :

$$h_1(k) = 123456 \bmod 701 = 80$$

$$h_2(k) = 1 + (123456 \bmod 700) = 1 + 256 = 257$$

1st probe : when $i = 1$

$$h(k, i) = h_1(k) + ih_2(k)$$

$$h(k, 1) = h_1(k) + h_2(k) = 80 + 257 = 337$$

2nd probe : when $i = 2$

$$h(k, 2) = h_1(k) + 2 \cdot h_2(k) = 80 + 2 \cdot 257$$

$$h(k, 2) = 80 + 514 = 594$$

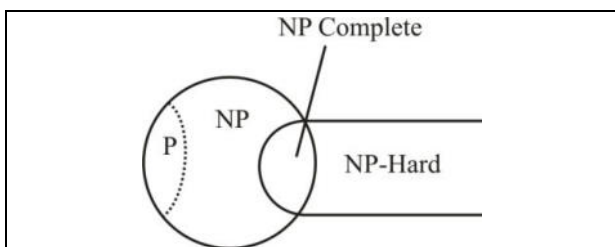
$$\text{So, difference between first two probes} = 594 - 337 = 257$$

67. Consider the complexity class CO – NP as the set of languages L such that $\bar{L} \in NP$, and the following two statements :
 $S_1 : P \subseteq CO - NP$
 $S_2 : \text{If } NP \neq CO - NP, \text{ then } P \neq NP$
 Which of the following is/are correct?
 (a) Only S_1
 (b) Only S_2
 (c) Both S_1 and S_2
 (d) Neither S_1 nor S_2

Ans. (c) : NP Complete problems :

They are those for which no polynomial time algorithm exists.

We can say a problem is NP complete if it is NP and belongs to NP hard.



NP problems :

A problem is a member of NP class if there exists a non-deterministic machine which can solve it in polynomial time.

NP Hard :

A problem is called as NP hard if all the NP class problems are polynomial time reducible to that and as hard as any problem of NP class.

68. Consider the following steps :

S_1 : Characterize the structure of an optimal solution

S_2 : Compute the value of an optimal solution in bottom-up fashion

Which of the step(s) is/are common to both dynamic programming and greedy algorithms?

- (a) Only S_1
- (b) Only S_2
- (c) Both S_1 and S_2
- (d) Neither S_1 nor S_2

Ans. (a) : Both Dynamic Programming and greedy algorithms find optimal substructure in the problem but only dynamic programming uses the bottom up approach.

69. Consider the following properties with respect to a flow network $G = (V, E)$ in which a flow is a real-valued function $f : V \times V \rightarrow \mathbb{R}$:

P_1 : For all $u, v \in V$, $f(u, v) = -f(v, u)$

P_2 : $\sum_{v \in V} f(u, v) = 0$ for all $u \in V$

Which one of the following is/are correct?

- (a) Only P_1
- (b) Only P_2
- (c) Both P_1 and P_2
- (d) Neither P_1 nor P_2

Ans. (a) : Flow for a real valued function $f : v \times v \rightarrow \mathbb{R}$ should satisfy following two properties :

- 1) skew symmetry property : i.e $f(u, v) = -f(v, u)$
- 2) capacity constraint means how should be less than the capacity of the It can not exceed the capacity i.e. $f(u, v) \leq c(u, v)$
- 3) Flow conservation : It means no vertex except the source and destination creates or store the flow.

Net flow should be 0.

i.e for all $u \in v - \{s, t\}$, $\sum_{v \in V} f(u, v) = 0$

- 70. Consider the following statements :**
- S₁ :** For any integer $n > 1$, $a^{\phi(n)} \equiv 1 \pmod{n}$ for all $a \in \mathbb{Z}_n$, where $\phi(n)$ is Euler's phi function.
- S₂ :** If p is prime, then $a^p \equiv 1 \pmod{p}$ for all $a \in \mathbb{Z}_p$
- Which one of the following is/are correct?**
- (a) Only S₁
 - (b) Only S₂
 - (c) Both S₁ and S₂
 - (d) Neither S₁ nor S₂

Ans. (a) : If m is a positive integer and a is an integer such that $(a, m) = 1$ then

$a^{\phi(m)} \equiv 1 \pmod{m}$ where $\phi(m)$ Euler's totient function

$$\phi(n) = \phi(P_1^{k_1}) \phi(P_2^{k_2}) \dots \phi(P_r^{k_r})$$

$$= P_1^{k_1} \left(1 - \frac{1}{P_1}\right) P_2^{k_2} \left(1 - \frac{1}{P_2}\right) \dots P_r^{k_r} \left(1 - \frac{1}{P_r}\right)$$

$$= P_1^{k_1} P_2^{k_2} \dots P_r^{k_r} \left(1 - \frac{1}{P_1}\right) \left(1 - \frac{1}{P_2}\right) \dots \left(1 - \frac{1}{P_r}\right)$$

$$= n \left(1 - \frac{1}{P_1}\right) \left(1 - \frac{1}{P_2}\right) \dots \left(1 - \frac{1}{P_r}\right)$$

Note for example

$$2^{\phi(9)} = 2^6 = 64 \equiv 1 \pmod{9}$$

So statement 1 is Euler's Theorem so S1 is correct.

Fermat's little theorem :→

states that if P is a prime number, then for any intezer, a the number $a^p - a$ is an integer multiple of P . In the notation of modular arithmetic, this is expressed as $a^p \equiv a \pmod{P}$.

$$a^{p-1} \equiv 1 \pmod{P}$$

for example, if $a = 2$ and $p = 7$, then $2^7 = 128$ and $128 - 2 = 126 = 7 \times 18$ is an integer multiple of 7.

So statement 2 is not fermat's little theorem S2 is false.

Or

Statement 1 : Correct Euler's theorem (Fermat - Euler theorem or Euler's totient theorem) states that if ' n ' and ' a ' are Coprime positive integers, than $a^{\phi(n)} \equiv 1 \pmod{n}$

Statement II - Incorrect

According to Fermat's Little theorem, Let ' P ' be a prime and ' a ' any integer, then $a^p \equiv a \pmod{p}$.

- 71. Which data structure is used by the compiler for managing variables and their attributes?**
- (a) Binary tree
 - (b) Link list
 - (c) Symbol table
 - (d) Parse table

Ans. (c) : Symbol table

It can be implemented by using an array, hash table, tree and even some time with the help of the linked list.

- 72. On translating the expression given below into quadruple representation, how many operations are required?**

$$(i * j) + (e + f) * (a * b + c)$$

- (a) 5
- (b) 6
- (c) 3
- (d) 7

Ans. (b) : 6 operation required

| | | | | |
|----|---|----|----|----|
| 1) | * | i | J | t1 |
| 2) | + | e | f | t2 |
| 3) | * | a | b | t3 |
| 4) | + | t3 | c | t4 |
| 5) | + | t1 | t2 | t5 |
| 6) | * | t5 | t4 | t6 |

73. Replacing the expression $4 * 2.14$ by 8.56 is known as

- (a) constant folding
- (b) induction variable
- (c) strength reduction
- (d) code reduction

Ans. (a) : Constant folding is compiler optimizations technique used by many modern compilers.

Constant folding is the process of recognizing and evaluation constant expressions at compile time rather than computing them at runtime

Example :

int var = $4 * 2.14$;

Replace with

int var = 8.56

Since multiplication is costly operation, it has been replaced constant which is more optimized version of previous code.

74. Shift-reduce parser consists of

- (a) input buffer
- (b) stack
- (c) parse table

Choose the correct option from those give below :

- (a) (a) and (b) only
- (b) (a) and (c) only
- (c) (c) only
- (d) (a), (b) and (c)

Ans. (d) : Shift reduce parser is based on the idea of predictive parsing with Look ahead. It is a bottom up parser.

Shift reduce parser generates a parse tree for an input string from bottom to top. At the end, reaches to the start symbol of the grammar. For a shift reduce parse, two data structure are required : one is stack and another in input buffer. Stack is used to keep the grammar symbols and input buffer hold the string to be parsed. Initially both the stack and input buffer contain the \$ symbol.

Example : $E \rightarrow E - E$ $E \rightarrow E \times E$ $E \rightarrow id$ we have to parse the string $id-id \times id$:

| Stack | Input Buffer | Parsing action |
|--------------------|------------------------|-----------------------------------|
| \$ | id - id $\times$ id \$ | shift |
| \$id | -id $\times$ id \$ | Reduce $E \rightarrow id$ |
| \$ E | -id $\times$ id \$ | shift |
| \$ E- | $\times$ id \$ | Reduce $E \rightarrow id$ |
| \$ E - E | $\times$ id | shift |
| \$ E - E \times | id \$ | shift |
| \$ E - E \times id | \$ | Reduce $E \rightarrow id$ |
| \$ E - E \times E | \$ | Reduce $E \rightarrow E \times E$ |
| \$ E - E | \$ | Reduce $E \rightarrow E - E$ |
| \$ E | \$ | Accept |

75. How many states are there in a minimum state automata equivalent to regular expression given below?

Regular expression is $a * b(a + b)$

(a) 1

(b) 2

(c) 3

(d) 4

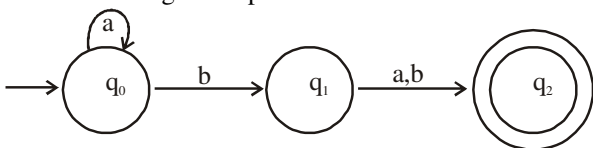
Ans. (c) : Regular expression : $a * b(a + b)$

Language that is accepted by this expression is of type :

{ba, bb, bab, aba, abb, abab, aabab,.....}

Smallest string length is 2, that is, we need at least 3 states for this.

NFA for this regular expression is :



So number of states required in this NFA are 3.

76. Match List-I with List-IIwhere L_1 : Regular language L_2 : Context-free language L_3 : Recursive language L_4 : Recursively enumerable language

| | List-Iq | | List-II |
|-----|------------------------|-------|---------------------------------|
| (A) | $\bar{L}_3 \cup L_4$ | (i) | Context-free language |
| (B) | $\bar{L}_2 \cup L_3$ | (ii) | Recursively enumerable language |
| (C) | $\bar{L}_1^* \cup L_2$ | (iii) | Recursive language |

Choose the correct option from those given below :

(a) (A)-(ii), (B)-(i), (C)-(iii)

(b) (A)-(ii), (B)-(iii), (C)-(i)

(c) (A)-(iii), (B)-(i), (C)-(ii)

(d) (A)-(i), (B)-(ii), (C)-(iii)

| Ans. (b) : | |
|----------------------|-------------------------------------|
| List-I | List-II |
| (a) $L_3^1 \cup L_4$ | (2) Recursively enumerable language |
| (b) $L_2^1 \cup L_3$ | (3) Recursive Language |
| (c) $L_1^* \cap L_2$ | (1) Context-free language |

Expression a : $L_3^1 \cup L_4$

L_3 is recursive language and L_4 is recursively enumerable language. Complement of recursive language is recursive. So, L_3^1 is also recursive. Also union of recursive and Recursively enumerable language. $L_3^1 \cup L_4$ is recursively enumerable.

Expression b : $L_2^1 \cup L_3$

L_2 is context free language. Context free Languages are not closed under complement. It is context sensitive. Context sensitive is also considered as recursive languages is also recursive.

Expression C : $L_1^* \cap L_2$

L_1 is regular language. Regular languages are closed under Kleene closure. So L_1^* is also regular. Intersection of a regular and context free language is a context free language. So, $L_1^* \cap L_2$ is a context free language.

77. How can the decision algorithm be constructed for deciding whether context-free language L is finite?

- (A) By constructing redundant CFG G in CNF generating language $\underline{L}$
- (B) By constructing non-redundant CFG G in CNF generating language L
- (C) By constructing non-redundant CFG G in CNF generating language $L - \{\epsilon\}$ (ϵ stands for null)

Which of the following is correct?

- (a) (A) only
- (b) (B) only
- (c) (C) only
- (d) None of (A), (B) and (C)

Ans. (c) : We can check that whether a context free language L is finite or not by constructing non redundant context free grammar in which there are no null productions no unit productions and no useless productions. Steps that are used in deciding whether a context free language is finite or not?

- 1) Construct the context free grammar corresponding to that language.
- 2) Reduce the grammar by removing null, unit and useless productions.
- 3) Construct the directed graph of that grammar
- 4) If that graph contains a cycle, then that language is infinite.

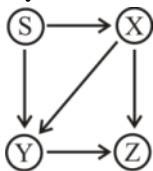
If graph does not contain cycle, it means language is finite.

$S \rightarrow XY/X$

$X \rightarrow YZ/Y$

$Y \rightarrow ZZ/Z$

This grammar is already reduced. Graph for this is



This graph does not contain any cycle, so Language is finite for this.

The decision algorithm be constructed for deciding whether context free language L is finite by constructing non redundant CFG G in CNF generating language $L - \{\wedge\}$ ($\wedge$ stands for null)

78. Consider the following grammar :

$S \rightarrow XY$

$X \rightarrow YaY \mid a$ and $Y \rightarrow bbX$

Which of the following statements is/are true about the above grammar?

- (A) Strings produced by the grammar can have consecutive three a's.
 - (B) Every string produced by the grammar have alternate a and b.
 - (C) Every string produced by the grammar have at least two a's.
 - (D) Every string produced by the grammar have b's in multiple of 2.
- (a) (A) only
 - (b) (B) and (C) only
 - (c) (D) only
 - (d) (C) and (D) only

Ans. (d) : $S \rightarrow XY$

$X \rightarrow YaY \mid a$ and $y \rightarrow bbX$

strings that are produced by this grammar are :

$S \rightarrow XY \rightarrow abbX \rightarrow abba$

$S \rightarrow XY \rightarrow YaYY$

$\rightarrow bbXabbXbbX \rightarrow bbaabbabba$

$S \rightarrow XY \rightarrow YaYY$

$\rightarrow bbXabbXbbX \rightarrow bbYaYabbXbbX$

$\rightarrow bbbbXabbXabbXbbX \rightarrow bbbbaabbaabbabba$

So, strings that are accepted by this language are in the form of :

$\{abba, bbaabbabba, bbbbaabbaabbabba, \dots\}$

smallest string contains two a and all other contains more than two a's It means every string must contains at least two a's. Also number of b's in the string is multiple of 2.

79. Which of the following problems is/are decidable problem(s) (recursively enumerable) on turning machine M ?

- (A) G is a CFG with $L(G) = \phi$

(B) There exist two TMs M_1 and M_2 such that $L(M) \subseteq \{L(M_1) \cup L(M_2)\}$ = language of all TMs

(C) M is a TM that accepts ω using at most $2^{|\omega|}$ cells of tape

- (a) (A) and (B) only
- (b) (A) only
- (c) (A), (B) and (C)
- (d) (C) only

Ans. (c) : a, b, c is correct option-

- a) $L(G) = \phi$ is a emptiness problem and emptiness problem for context free languages is decidable.
- b) Recursive : The reason is we can assume $L(m_2) = \Sigma^*$ and hence any $TM(m_1)$ will be definitely subset of $L(m_2)$. So we only need to see that M_1 is a valid turing machine and no need to check whether it is subset of $L(m_2)$ as it is obviously subset of $L(m_2) = \Sigma^*$ checking validity of turing machine is always a halting process, hence it is recursive.

80. For a statement

A language $L \subseteq \Sigma^+$ is recursive if there exists some turing machine M.

Which of the following conditions is satisfied for any string ω ?

- (a) If $\omega \in L$, then M accepts ω and M will not halt
- (b) If $\omega \notin L$, then M accepts ω and M will not halt
- (c) If $\omega \notin L$, the M halts without reaching to acceptable state
- (d) If $\omega \in L$, then M halts without reaching to an acceptable state

Ans. (c) : A language is recursive if there exists a Turing machine. For strings that are in the language, Turing machine accepts the language and always halts and for strings not in the language, string will halt in the non final state.

Explanation :-

A language $L \subseteq \Sigma^*$ is recursive if there exists some Turing machine M suppose w is the string.

So if w belongs to language, then Turing machine M will halt by reaching at final state and if does not belong to L, then M halts without reaching to final state (acceptable state).

81. A fully connected network topology is a topology in which there is a direct link between all pairs of nodes. Given a fully connected network with n nodes, the number of direct links as a function of n can be expressed as

- (a) $\frac{n(n+1)}{2}$
- (b) $\frac{(n+1)}{2}$

(c) $\frac{n}{2}$

(d) $\frac{n(n-1)}{2}$

Ans. (d) : If there are n nodes then we can connect node 1 to $n-1$ nodes using $n-1$ direct links



We can connect node 2 to $n-1$ nodes using $n-2$ direct links (not $n-1$ since we have already connected it with node 1)

We can connect node 3 to $n-1$ nodes using $n-3$ direct links (not $n-1$ since we have already connected it with node 1 and node 2)....

We can connect node n to $n-1$ nodes using 0 direct links (since it is already connected to every other node)

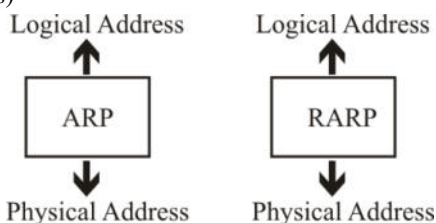
So total number of links required = $(n-1) + (n-2) + (n-3) + \dots + 1 + 0 = \frac{n(n-1)}{2}$

82. What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?

- (a) ARP
- (b) DNS
- (c) RARP
- (d) ICMP

Ans. (c) : What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?

Reverse Address Resolution Protocol - It is used to get IP address with the help of physical address (MAC Address)



ARP (Address resolution protocol) is a protocol used by the Internet protocol (IP), specifically IPV4, to map IP network addresses to the hardware addresses used by a data link protocol, that is MAC address.

DNS (Domain Name System) DNS is a host name to IP address translation service. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers.

RARP :- The Reverse Address Resolution Protocol (RARP) is a computer networking protocol used by a

client computer to request its Internet protocol (IP) address from a computer network, when all it has available is its link layer or hardware address, such as a MAC address. RARP protocol broadcast message with its MAC address and receive an IP address in reply. The Internet Control Message Protocol (ICMP)- The internet control Message protocol (ICMP) is a supporting protocol in the Internet protocol suite. It is used by network devices, including routers, to send error messages and operational information indicating success or failure when communicating with another Host IP address.

- 83. You are designing a link layer protocol for a link with bandwidth of 1 Gbps (10^9 bits/second) over a fiber link with length of 800km. Assume the speed of light in this medium is 200000 km/second. What is the propagation delay in this link?**
- (a) 1 millisecond
(b) 2 milliseconds
(c) 3 milliseconds
(d) 4 milliseconds

Ans. (d) : Data

Bandwidth (BW) = 1Gbps = 10^{10} b/s

Length (d) = 800 km

speed (v) = 200000 km/s = 2×10^5 km/s

propagation delay = T_p

Method 1

Formula :

$$\{\{\rm{T}\} - \{\rm{P}\}\} = \frac{\{\{\rm{d}\}\}}{\{\{\rm{v}\}\}}$$

Calculation :

$$\{\{\rm{T}\} - \{\rm{P}\}\} = \frac{\{800\{\rm{;km}\}\}}{\{2 \times 10^5\}}$$

- 85. Consider the following two statements with respect to IPv4:**

P : The loopback (IP) address is a member of class A

Q : The loopback (IP) address is used to send a packet to the local host

What can you say about the statements P and Q?

- (a) P-True; Q-False (b) P-False; Q-True
(c) P-True; Q-True (d) P-False; Q-False

Ans. (b) : IPV4 address : 32 bits

IP address = Network ID + Host ID

Network ID + Host ID = 32

| class | PREFIX
(bits reserved
from I st byte) | Network
ID (bits) | Range of I st
byte | No. of
addresses
consumed |
|---------|--|----------------------|----------------------------------|---------------------------------|
| class A | 0 | 8 | 0-127 | 2^3 |
| class B | 10 | 16 | 128-191 | 2^3 |
| class C | 110 | 24 | 192-223 | 2^2 |
| class D | 1110 | — | 224-239 | 2^2 |
| class E | 1111 | — | 240-255 | 2^2 |

The address between the range 127.0.0.0 to 127.255.255 is reserved and used as a loopback address that falls in the range of class A. Therefore, statement P is false and statement Q is true.

$$\frac{200000 \text{ km}}{S} \backslash \left(\frac{T - P}{10^{-3}} \right) = \frac{1000 \times 250}{S}$$

$$T_p = 4 \text{ milliseconds.}$$
 Method 2 - propagation delay = distance / speed of the medium $\neq d/s$
 propagation delay = 800 km/200000 km/sec
 = (800/200000) sec
 = 4 milliseconds.

84. You need 500 subnets, each with about 100 usable host addresses per subnet. What network mask will you assign using a class B network address?
 (a) 255.255.255.252
 (b) 255.255.255.128
 (c) 255.255.255.0
 (d) 255.255.254.0

Ans. (b) :
 * In class B addressing : 16 - bit network ID and 16-bits for host ID.
 * 500 subnets are required and $2^9 = 512$, hence 9 bits are needed from host id part of class B addressing in order to provide 500 subnets.
 * Remaining 7 bits numbers of host possible per subnet is $128 - 2 = 126$
 subnet - id (25 bits)/ Host - id (7-bits)
 Subnet mask contains is in subnet id part and the os in host ID part. Hence subnet mask is :
 11111111.11111111.11111111.10000000
 In decimal form it can be represented as :
 255.255.255.128

et to IPv4 in computer networking :

B network.

ket from host to itself.

?

| of Ip
ess
med | Number of
networks | IP address
per network | Per lentage of
IPV4 address
consumed |
|---------------------|-----------------------|---------------------------|--|
| | 2^7 | 2^{24} | 50% |
| | 2^{14} | 2^{16} | 25% |
| | 2^{21} | 2^8 | 12.5% |
| | Multicast Reserved | | 6.25% |
| | | | 6.25% |

5.255 (exclude 127.0.0.0 and 127.255.255.255) can be
ss A network. Therefore, the loopback (IP) address is a
e.

A loopback address is primarily used to validate that the network is properly configured and the TCP/IP stack installed. A data packet is sent to the loopback address and is sent back to the source application (to itself). Therefore, statement Q is true.

86. In the TCP/IP model, encryption and decryption are functions of _____ layer.

- (a) data link
- (b) network
- (c) transport
- (d) application

Ans. (d) : TCP/IP model

| Layers | Functions |
|------------------------|--|
| Application Layer | Authentication, Authorization, Encryption, Decryption etc. |
| Transport Layer | End to end connectivity segmentation, Multiplexing, demultiplexing, congestion control etc |
| Internet Layer | Host to host connectivity fragmentation etc |
| Link layer (data link) | flow control, framing etc |

87. Which of the following statements is/are true with regard to various layers in the Internet stack?

P : At the link layer, a packet of transmitted information is called a frame.

Q : At the network layer, a packet of transmitted information is called a segment.

- (a) P only
- (b) Q only
- (c) P and Q
- (d) Neither P nor Q

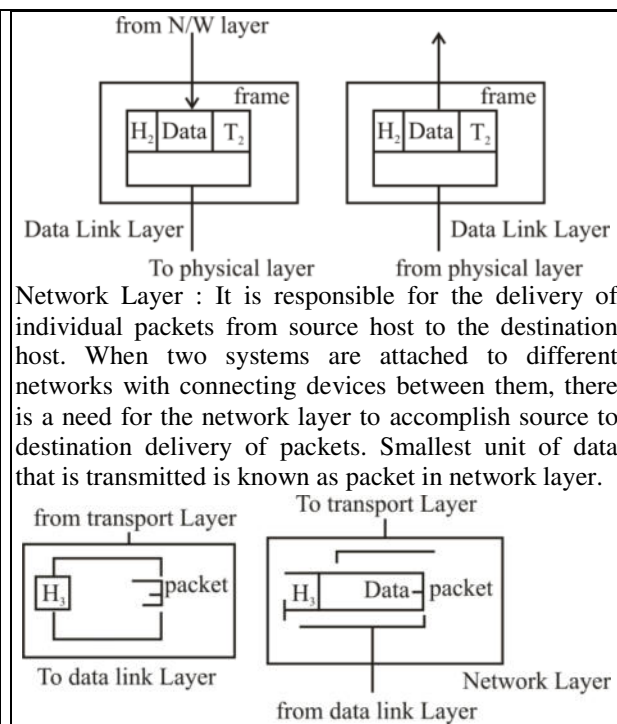
Ans. (a) : OSI model is a Layered framework for the design of network systems that allows communication between all types of computer systems. It consists of seven layers but related layer each of which defines a part of the process of moving information across a network.

The seven layers from bottom to top are as follows : physical layer, data link layer, network layer, transport layer, session layer, presentation layer, application layer.

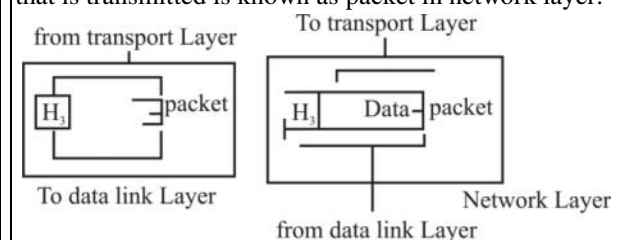
But here we have to focus only on data link layer and network layer.

Data link layer : This layer is responsible for moving frames from one hop to next. Data link layer divides the stream of bits received from the network layer into manageable units called frames. It adds a header to the frame to define the sender and receiver of the frame. It provides a flow and error control mechanism.

the locally connected physical network card is working present on a loopback address, never leaves the host system



Network Layer : It is responsible for the delivery of individual packets from source host to the destination host. When two systems are attached to different networks with connecting devices between them, there is a need for the network layer to accomplish source to destination delivery of packets. Smallest unit of data that is transmitted is known as packet in network layer.



88. What percentage (%) of the IPv4, IP address space do all class C addresses consume?

- (a) 12.5% (b) 25%
(c) 37.5% (d) 50%

Ans. (a) :

| | | | 21 Bit | 8 bit |
|---|---|---|---------|-------|
| 1 | 1 | 0 | Network | Host |
| ----- Class C----- | | | | |
| - | | | | |
| Total No of IPV4 addresses possible = 2^{32} | | | | |
| Total No of IPV4 address belong to class C = 2^{29} | | | | |
| [As class C IP address always start with 110] | | | | |
| So required percentage = $(2^{29}/2^{32}) \times 100$ | | | | |
| = 100/8 | | | | |
| = 12.5% | | | | |

89. The ability to inject packets into the Internet with a false source address is known as

- (a) man-in-the-middle attack
(b) IP phishing
(c) IP sniffing
(d) IP spoofing

Ans. (d) : IP spoofing :-

IP address spoofing or IP spoofing is the creation of Internet protocol packets with a false source IP address, for the purpose of impersonating another computing system.

Main-in-the-Middle Attack : It is an attack where the attacker secretly relays and possibly alters the

communications between two parties who believe they are directly communicating with each other three security issue that are particularly applicable to the IP protocol : **packet sniffing, packet modification, and IP spoofing.**

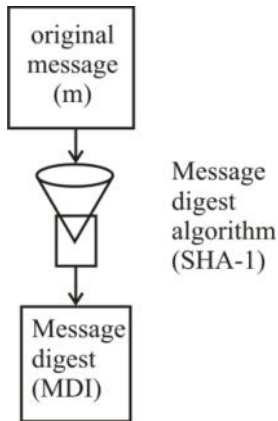
Packet sniffing : In packet sniffing, an intruder tries to intercept the IP packet and make a copy of it It is a passive attack, in which the attacker does not change the contents of the packet.

IP spoofing : In IP spoofing, an attacker can masquerade (Pretense) somebody else and creates an IP packet that carries the source address of another computer. An attacker can send an IP packet to a bank pretending that is coming from one of the customers.

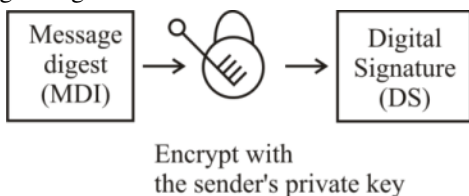
90. The RSA encrypting algorithm also works in reverse, that is, you can encrypt a message with the private key and decrypt it using the public key. This property is used in
- (a) intrusion detection systems
 - (b) digital signatures
 - (c) data compression
 - (d) certification

Ans. (b) : RSA encryption algorithm when works in reverse, it uses digital signature technique. Step by step method of this is as follows:

- 1- Sender (A) uses the SHA-1 message digest method to calculate the message digest over the original message (m).

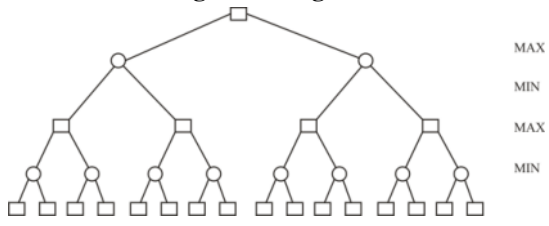


- 2- Sender now encrypts the message digest with his/her private key and that output is called as digital signature of sender.



- 3- Now, sender sends the original message along with digital signature to receiver B.
- 4- After that, when it reaches to the receiver, receiver decrypts this using the sender's public key.

91. Consider the game tree given below :

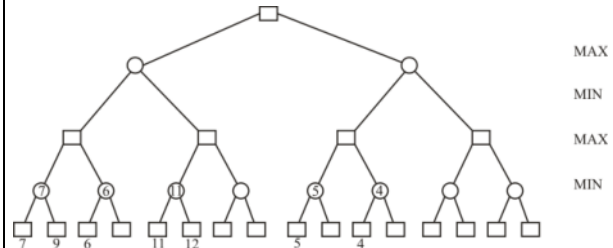


Here ○ and □ represent MIN and MAX nodes respectively. The value of the root node of the game tree is

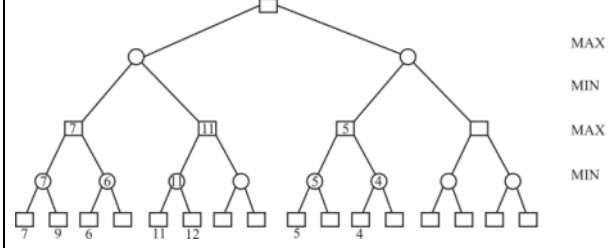
- (a) 4 (b) 7
(c) 11 (d) 12

Ans. (b) : Start from bottom to top :

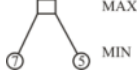
1) First it finds the minimum value from the given values.



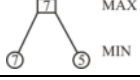
2) At next step, max value is considered.



3) Then again min value is considered



4) The at the root, we have to consider the maximum value



92. Match List-I with List-II

| | | | |
|------------|--------------------------------|--------------|--------------------------------|
| | List-I | | List-II |
| (A) | Greedy best-first | (i) | Minimal cost (p) + h(p) |
| (B) | Lowest cost-first | (ii) | Minimal h(p) |
| (C) | A⁺ algorithm | (iii) | Minimal cost (p) |

Choose the correct option from those given below :

- (a) (A)-(i), (B)-(ii), (C)-(iii)
 (b) (A)-(iii), (B)-(ii), (C)-(i)
 (c) (A)-(i), (B)-(iii), (C)-(ii)
 (d) (A)-(ii), (B)-(iii), (C)-(i)

Ans. (d) :

- a) Greedy best first → ii) Minimal $h(p)$
 - b) Lowest cost → iii) Minimal cost (p)
 - c) A* search → i) Minimal cost $(p) + h(p)$
- $h(p)$ = heuristic cost from current node to goal node.
Cost (p) = Cost to reach that node.

93. The STRIPS representation is

- (a) a feature-centric representation
- (b) an action-centric representation
- (c) a combination of feature-centric and action-centric representations
- (d) a hierarchical feature-centric representation

Ans. (b) : The STRIPS representation is-

STRIPS - It stands for stanford research institute problem solver. It is an action centric representation in which for each action, it specifies the effect of the action.

In strips representation, there are states and goal and operators.

operator is of three parts:

- 1- Operator name
- 2- Precondition : it must occur so that operator can execute.
- 3- Effect : it means how the state will change after an action.

In STRIPS, there is state base view of time. Actions are external to the logic. Given a state and an action, STRIPS representation is used to find:

- a) Whether the action can be carried out in the state.
- b) what is true in the resulting state. Predicates used in STRIPS are primitives and derived. It is used to determine the truth value of primitive predicates based on the previous state and action.

94. A fuzzy conjunction operator denoted as $t(x, y)$ and a fuzzy disjunction operator denoted as $s(x, y)$ form a dual pair if they satisfy the condition :

- (a) $t(x, y) = 1 - s(x, y)$
- (b) $t(x, y) = s(1 - x, 1 - y)$
- (c) $t(x, y) = 1 - s(1 - x, 1 - y)$
- (d) $t(x, y) = s(1 + x, 1 + y)$

Ans. (c) : $t(x, y) = 1 - s(1 - x, 1 - y)$

FuzzySet :A fuzzy set is that which allows its members to have different value in the set interval $[0, 1]$

Operations that can be performed on fuzzy set are union, intersection, complement.

Union and intersection operation in fuzzy set are equal to conjunction (AND) and disjunction operation (or) respectively.

We can consider min for conjunction and max for disjunction. Also, there are various other choices also. Such as to calculate product for conjunction and algebraic sum for disjunction operation.

If fuzzy conjunction operator is denoted as $f(x, y)$ and fuzzy disjunction operator is denoted as $s(x, y)$. Then it forms as dual pair if : $1 - t(x, y) = S(1 - x, 1 - y)$
or $1 - s(x, y) = t(1 - x, 1 - y)$
Here, it means choice of fuzzy disjunction operator determines the choice of fuzzy conjunction operator and vice-versa.

95. Let A_{α_0} denotes the α – cut of a fuzzy set A at α_0 . If $\alpha_1 < \alpha_2$, then

- (a) $A_{\alpha_1} \supseteq A_{\alpha_2}$
- (b) $A_{\alpha_1} \supset A_{\alpha_2}$
- (c) $A_{\alpha_1} \subseteq A_{\alpha_2}$
- (d) $A_{\alpha_1} \subset A_{\alpha_2}$

Ans. (a) : Alpha cut in a fuzzy set :

An α cut of a fuzzy set A is a crisp set that contains all elements in U that have membership value in A greater than or equal to α .

Explanation :

$A_{\alpha_0} = \alpha$ – cut of a fuzzy set A at α_0

if we take $\alpha_1 < \alpha_2$

consider an example for this :

Fuzzy set A on interval $x[1, 5]$

membership function (U_A) is $= (x)/(x + 2)$

Case 1 : first find α 0.2

x can be 1, 2, 3, 4 and 5

when $x = 1$, $u_A = 1/1 + 2 = 1/3 = 0.33$

when $x = 2$ $u_A = 2/2 + 2 = 1/2 = 0.5$

when $x = 4$ $u_A = 4/4 + 2 = 4/6 = 0.66$

when $x = 5$ $u_A = 5/7 = 0.71$

Alpha cut for $\alpha = 0.2$ is $\{1, 2, 3, 4, 5\}$

Case 2- Take α 0.5

Alpha cut for this set is $\{2, 3, 4, 5\}$

from here it is clear that set 2 is subset of set 1 i.e $A_{\alpha_1} \supseteq A_{\alpha_2}$

96. Consider the following methods :

M_1 : Mean of maximum

M_2 : Centre of area

M_3 : Height method

Which of the following is/are defuzzification method(s)?

- (a) Only M_2
- (b) Only M_1 and M_2
- (c) Only M_2 and M_3
- (d) M_1 , M_2 and M_3

Ans. (d) : Defuzzification: It is the reverse of fuzzification in which a fuzzy set is transformed back to crisp set.

various defuzzification methods are

- 1) centre of sum method: In this method, it covers the area two times, which is the required area.
- 2) Centroid method: It provides crisp value by considering the centre of gravity on fuzzy set. It is also known as centre of gravity method. Total area is divided into sub areas in this. Centre of gravity is calculated for each sub area.

- 3) Centre of area method: It calculates the area under curve. It partitions a region into two regions of same area.
- 4) weight average method: In this membership functions are used. Each membership function is weighted by the maximum membership value.
- 5) Maxima methods : a) First of maxima: It finds the smallest value of domain with maximum membership value.
b) Last of maxima: It finds the largest value with maximum membership value.
c) Mean of maxima: In this, mean of the values is taken with maximum membership value.

97. Consider the following :

(A) Evolution

(B) Selection

(C) Reproduction

(D) Mutation

Which of the following are found in genetic algorithms?

- (a) (A), (C) and (D) only
- (b) (B) and (D) only
- (c) (A), (B), (C) and (D)
- (d) (A), (B) and (D) only

Ans. (c) : Genetic algorithm are adaptive methods which may be used to solve search and optimization problems. They are based on the genetic process of biological organisms.

Explanation: Genetic algorithm use a direct analogy of natural behavior. They work with a population of individuals, each representing a possible solution to a given problem. Each individual is assigned a fitness score according to how good a solution to the problem it is. The highly fit individual are given opportunities to reproduce, by cross breeding. This produces offspring. Before a GA can be run, a suitable coding and fitness function is required.

Phases that are considered in genetic algorithm are:

- 1) Initialize population
- 2) Selection
- 3) Crossover
- 4) Mutation
- 5) reproduction

98. Which of the following is an example of unsupervised neural network?

- (a) Back-propagation network
- (b) Hebb network
- (c) Associative memory network
- (d) Self-organizing feature map

Ans. (d) : A Self-Organizing Feature Map: It is a type of artificial neural network (ANN) that is trained using unsupervised learning to produce a low-dimensional (typically two-dimensional), discretized representation of the input space of the training samples, called a map, and is therefore a method to do dimensionality reduction.

Self-organizing maps differ from other artificial neural network as opposed to error correction learning (such as back propagation with gradient descent), and in the sense that they use a neighborhood function to preserve the topological properties of the input space.

99. The value of the derivative of Sigmoid function

given by $f(x) = \frac{1}{1+e^{-2x}}$

at $x=0$ is

(a) 0

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) ∞

Ans. (b) :

$$\begin{aligned} & \frac{d}{dx} \left(\frac{1}{1+e^{-2x}} \right) \\ &= \frac{(1+e^{-2x}) \frac{d}{dx}(1) - (1) \frac{d}{dx}(1+e^{-2x})}{(1+e^{-2x})^2} \\ &= \frac{0 - (1) \frac{d}{dx}(1+e^{-2x})}{(1+e^{-2x})^2} = \frac{-(1)(e^{-2x})(-2)}{(1+e^{-2x})^2} \\ &= \frac{2e^{-2x}}{(1+e^{-2x})^2} \end{aligned}$$

Now put $x = 0$

$$\frac{2e^{-2(0)}}{(1+e^{-2(0)})^2} = \frac{2(1)}{(1+1)^2} = \frac{2}{4} = \frac{1}{2}$$

100. Reinforcement learning can be formalized in terms of ____ in which the agent initially only knows the set of possible ____ and the set of possible actions.

(a) Markov decision processes, objects

(b) Hidden states, objects

(c) Markov decision processes, states

(d) objects, states

Ans. (c) : Reinforcement learning can be formalized in terms of Markov decision processes, in which the agent initially only knows the set of possible states and the set of possible actions.